

A review of ‘Learning of state-space models with highly informative observations: a tempered Sequential Monte Carlo solution’

Author: Finn van Vlaanderen

MSc Mathematics, Vrije Universiteit Amsterdam, Seminar Sequential Monte Carlo Methods, taught by Frank van der Meulen

May 27, 2026

1. Introduction

Given some state-space model with trajectories dependent upon some parameter, an interesting problem is that of Bayesian parameter estimation provided some observed trajectory from the state-space model. This has been discussed extensively in general (c.f. Kantas et al., 2015), but in this text we will regard the particularly challenging case where the observations are *highly informative*, that is, where the variability in the observations is very small relative to the variability in the latent space trajectories. Specifically, we will review the monograph by Svensson et al., 2018, where the authors propose a Sequential Monte Carlo-based algorithm that intends to solve this problem. We will reformulate the algorithms proposed in Svensson et al., 2018 in generalized terms using the language of Feynman-Kac models, following the conventions used by Chopin and Papaspiliopoulos, 2020. Furthermore, we will showcase a reproduction of some of their results, provide supplementary analyses, and briefly discuss their implications.

2. The problems with highly informative observations

Before specializing to the more limited range of state-space models discussed in Svensson et al., 2018 and directly amenable to their solution, let us consider more generally state-space models of the form

$$\begin{aligned} x_0 &\sim P_0^\theta(dx_0), \\ x_t|x_{t-1} &\sim P_t^\theta(dx_t, x_{t-1}), & t = 1, \dots, T \\ y_t|x_t &\sim F_t^\theta(dy_t, x_t), & t = 0, \dots, T \end{aligned}$$

where P_t^θ, F_t^θ are known Markov kernels defined on a shared measurable space $\mathcal{X}$ and depending measurably on some unknown parameter $\theta \in \Theta$. When the variance of $y_t|x_t$, which we will call the *measurement noise*, is much smaller than the variance of $x_t|x_{t-1}$, which we will call the *process noise*, we say that this state-space model has highly-informative observations¹. An extreme example would be when $F_t^\theta(dy_t, x_t) = \delta_{x_t}(dy_t)$; when the observations depend deterministically on the latent states. For measure-theoretic ease and continuity with Chopin and Papaspiliopoulos, 2020, however, we will assume throughout the remainder of this section that the kernels F_t^θ admit densities $f_t^\theta(\cdot|x_t)$. Deterministic observations can still be approximated by, for example, taking $f_t^\theta(y_t|x_t) = N(y_t|x_t, \sigma^2)$ with negligible σ^2 .

To learn the parameters $\theta \in \Theta$ from given observations $y_{0:T} \in \mathcal{X}^{T+1}$, one can adopt a Bayesian approach by defining a prior $\nu(d\theta)$

on Θ and estimating the posterior $\mathbb{P}_T(d\theta|Y_{0:T} = y_{0:T})$. An application of Bayes’ theorem to the joint posterior $\theta, x_{0:T}|y_{0:T}$ combined with marginalization over $x_{0:T}$ yields

$$\mathbb{P}_T(d\theta|Y_{0:T} = y_{0:T}) = \frac{p_T^\theta(y_{0:T})\nu(d\theta)}{\int p_T^\theta(y_{0:T})\nu(d\theta)},$$

where $p_T^\theta(y_{0:T})$ is the likelihood of the state-space model, which exists because F_T^θ admits a density. If the likelihoods are known beforehand, procedures like Markov Chain Monte Carlo sampling can be applied directly to sample from this posterior, hence solving the parameter learning problem.

This assumption is not easily met. In state-space models with non-linear latent trajectories, the likelihood is usually intractable (Svensson et al., 2018). In this case, an unbiased estimate of the likelihood can still be obtained by using a particle filter (see Algorithm 2.1 for reference).

Algorithm 2.1: Generic particle filter algorithm

Input: Feynman-Kac model $\mathbb{M}_0, M_{1:T}, G_{0:T}$
Output: Particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$
 Run for $n = 1, \dots, N$

```

 $X_0^n \sim \mathbb{M}_0(dx_0)$ 
 $w_0^n \leftarrow G_0(X_0^n)$ 
 $W_0^n \leftarrow w_0^n / \sum_{m=1}^N w_0^m$ 
for  $t = 1$  to  $T$  do
   $A_t^{1:N} \sim \text{resample}(W_{t-1}^{1:N})$ 
   $X_t^n \sim M_t(X_{t-1}^{A_t^n}, dx_t)$ 
   $w_t^n \leftarrow G_t(X_{t-1}^{A_t^n}, X_t^n)$ 
   $W_t^n \leftarrow w_t^n / \sum_{m=1}^N w_t^m$ 
end for
  
```

In the bootstrap formalism corresponding to the state-space model under consideration, we would select the Feynman-Kac model with initial distribution $\mathbb{M}_0^\theta(dx_0) = P_0^\theta(dx_0)$, transition kernels $M_t^\theta(dx_t, x_{t-1}) = P_t^\theta(dx_t, x_{t-1})$, and potential functions $G_0^\theta(x_0) = f_0^\theta(y_0|x_0)$, $G_t^\theta(x_t, x_{t-1}) = f_t^\theta(y_t|x_t)$. The normalizing constants L_t^θ in this Feynman-Kac model correspond to the likelihoods, and an unbiased estimate is given in general by

$$L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) = \left(\frac{1}{N} \sum_{n=1}^N w_0^n \right) \prod_{t=1}^T \left(\frac{1}{N} \sum_{n=1}^N w_t^n \right),$$

where $w_0^n = G_0^\theta(X_0^n)$ and $w_t^n = G_t^\theta(X_{t-1}^{A_t^n}, X_t^n)$. For a derivation, see Chopin and Papaspiliopoulos, 2020. In our case, this reduces to

¹One could be more careful here regarding the conditioning and what is meant by ‘much smaller’, but we omit this rigor and opt (like Svensson et al., 2018) for a heuristic characterization of highly-informativeness.

the estimate

$$p_T^\theta(y_{0:T}) \approx \prod_{t=0}^T \left(\frac{1}{N} \sum_{n=1}^N f_t^\theta(y_t | X_t^n) \right).$$

Let us furthermore write $Z = (X_{0:T}^{1:N}, A_{1:T}^{1:N})$ as a random variable considered to be drawn from the kernel $\psi_T^\theta(dz)$, where ψ_T^θ denotes the probability law induced by Algorithm 2.1 on the space $\mathcal{Z} := \mathcal{X}^{N(T+1)} \times \mathcal{A}^{NT}$, with $\mathcal{A}$ being the space in which the indices A_t^n take their values. This notational convention will be useful later on.

In principle, such estimates could be used within Markov Chain Monte Carlo to estimate likelihoods on the fly. One such MCMC method incorporating these likelihood estimates is Particle Marginal Metropolis-Hastings (PMMH), which we will touch upon shortly. This approach runs into major issues when observations are highly informative (Svensson et al., 2018), however, and cannot solve the parameter learning problem on its own. The central problem is that of *weight degeneracy*: when observation densities are very sharply peaked relative to the process noise, the mappings $x_t \mapsto f_t^\theta(y_t | x_t)$ assign non-negligible mass only to a very small region of the state space $\mathcal{X}$. Consequently, weights will tend to collapse and the likelihood estimator $L_T^N(\theta, Z)$ will exhibit large variance, often growing rapidly with the time horizon T . Furthermore, this problem is exacerbated when the parameter θ used to run the particle filter is far from the true parameter with which the observations were generated.

Svensson et al., 2018 work around these limitations by nesting PMMH kernels within a tempered Sequential Monte Carlo sampler; the PMMH kernels handle approximate sampling from intractable posteriors, and the tempering sequence handles the variance in the PMMH likelihood estimates. Let us now briefly explain the background behind these two components before moving on to Svensson et al., 2018's complete methodology.

3. Metropolis-Hastings for intractable likelihoods

The Particle Marginal Metropolis-Hastings algorithm, designed to allow sampling from posteriors with intractable likelihoods, differs from ordinary Metropolis-Hastings in that the likelihood of the target distribution is replaced by its unbiased particle filter estimate.

A PMMH sampler is equivalent to an ordinary Metropolis-Hastings sampler on the extended state space $\Theta \times \mathcal{Z}$, having proposal kernel $Q(d\tilde{\theta}, \theta) \psi_T^\theta(dz)$ and stationary distribution

$$\pi(d\theta, dz) = \frac{\nu(d\theta) L_T^N(\theta, z) \psi_T^\theta(dz)}{\int_{\Theta \times \mathcal{Z}} \nu(d\theta) L_T^N(\theta, z) \psi_T^\theta(dz)}.$$

The true posterior $\mathbb{P}_T(d\theta | Y_{0:T} = y_{0:T})$ is the marginal of the actual stationary distribution $\pi(d\theta, dz)$, which can be seen as follows:

$$\begin{aligned} \int_{z \in \mathcal{Z}} \pi(d\theta, dz) &\propto \nu(d\theta) \int_{\mathcal{Z}} L_T^N(\theta, z) \psi_T^\theta(dz) \\ &= \nu(d\theta) \mathbb{E}_{z \sim \psi_T^\theta} L_T^N(\theta, z) \\ &= \nu(d\theta) p_T^\theta(y_{0:T}). \end{aligned}$$

For a detailed treatment, we refer to Chapter 16 in Chopin and Papaspiliopoulos, 2020.

Algorithm 3.1: Particle Marginal Metropolis-Hastings (PMMH) sampler (one step)

Input: initial parameter θ , corresponding particle trajectories $(X_{0:T}^{1:N}, A_{1:T}^{1:N})$

Output: next parameter and particle trajectories

$\tilde{\theta} \sim Q(d\tilde{\theta}, \theta)$

Run particle filter to generate variables $(\tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N})$ given proposed parameter $\tilde{\theta}$

$U \sim \text{Unif}([0, 1])$

$\alpha \leftarrow \frac{\nu(\tilde{\theta}) L_T^N(\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}) q(\theta | \tilde{\theta})}{\nu(\theta) L_T^N(\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}) q(\tilde{\theta} | \theta)}$

if $U \leq \min(\alpha, 1)$ **then**

return $\tilde{\theta}, \tilde{X}_{0:T}^{1:N}, \tilde{A}_{1:T}^{1:N}$

else

return $\theta, X_{0:T}^{1:N}, A_{1:T}^{1:N}$

end if

4. Tempering for highly informative observations

Svensson et al., 2018 specialize their solution to the case where the target state-space model is of the form

$$\begin{aligned} x_0 &\sim P_0^\theta(dx_0), \\ x_t | x_{t-1} &\sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T \\ y_t | x_t &\sim N(g^\theta(x_t), \sigma^2), \quad t = 0, \dots, T, \end{aligned}$$

where σ^2 is negligibly small relative to the process noise, $g^\theta: \mathcal{X} \rightarrow \mathcal{Y}$ is some parameter-dependent deterministic function of x_t and the latent space dynamics are governed by a single kernel P^θ . Degenerate models can be approximated by setting $\sigma^2 \approx 0$.

To mitigate weight degeneracy in the PMMH approximations of the posterior, Svensson et al., 2018 employ a tempering strategy. The idea behind tempering is roughly as follows: given a sequence $\lambda_0 > \dots > \lambda_K \approx 0$, define a family of tempered posteriors $\{\pi(d\theta, dz | \lambda_k)\}_{k=0}^K$ interpolating between a model that is easy to sample from and the target posterior. Samples are then propagated through this sequence using an SMC sampler.

The particular tempering scheme proposed by Svensson et al., 2018 is as follows: given some decreasing sequence $(\lambda_k)_{k=0}^K$, define $\pi(d\theta, dz | \lambda_k)$ to be the posterior on the extended parameter space in a modified state-space model

$$\begin{aligned} x_0 &\sim P_0^\theta(dx_0), \\ x_t | x_{t-1} &\sim P^\theta(dx_t, x_{t-1}), \quad t = 1, \dots, T \\ y_t | x_t &\sim N(g^\theta(x_t), \sigma^2 + \lambda_k), \quad t = 0, \dots, T. \end{aligned}$$

Thus, when λ_k is sufficiently large, observations will no longer be highly informative according to the state-space model assumed in particle filtering, allowing reliable estimation of likelihoods and posteriors. When $\lambda_k = 0$, the target distribution is recovered.

The tempering scheme described above naturally induces an SMC sampler on the extended space $\Theta \times \mathcal{Z}$, which we now formulate in the language of Feynman-Kac models. We should note that the construction that follows is similar, but not identical, to the construction of

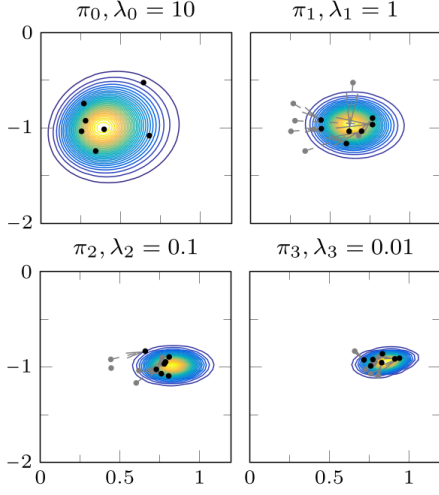


Figure 1: Contour plots of consecutive tempered likelihoods. The likelihoods plotted here are the true likelihoods, not their PMMH estimates. Black dots represent samples, propagated across distributions using Sequential Monte Carlo. Image from Svensson et al., 2018.

the SMC² sampler (see Chopin and Papaspiliopoulos, 2020, Chapter 18, and Chopin, Jacob, et al., 2012). Let $M(d\theta_k, dz_k, \theta_{k-1}, z_{k-1} | \lambda_k)$ denote the Markov kernel induced by Particle Marginal Metropolis-Hastings when the likelihood estimates are based on the tempered state-space model with temperature λ_k . The tempered posterior $\pi(d\theta, dz | \lambda_k)$ is the invariant distribution of this kernel, and hence we can use it as a mutation kernel in the construction of an SMC sampler. The distributions in the sequence of tempered posteriors $\{\pi(d\theta, dz | \lambda_k)\}_{k=0}^K$ can be written as

$$\pi(d\theta, dz | \lambda_k) = \nu(d\theta) \frac{L_T^N(\theta, z | \lambda_k) \psi_T^\theta(dz | \lambda_k)}{L_k}.$$

Assuming that the Radon-Nikodym derivative $\frac{\psi_T^\theta(dz | \lambda_k)}{\psi_T^\theta(dz | \lambda_{k-1})}$ exists, consecutive tempered posteriors can then be related to each other via

$$\begin{aligned} \pi(d\theta_k, dz_k | \lambda_k) &= \frac{L_{k-1}}{L_k} \frac{L_T^N(\theta_k, z_k | \lambda_k) \psi_T^\theta(dz_k | \lambda_k)}{L_T^N(\theta_{k-1}, z_{k-1} | \lambda_{k-1}) \psi_T^\theta(dz_{k-1} | \lambda_{k-1})} \pi(d\theta_{k-1}, dz_{k-1} | \lambda_{k-1}) \\ &= \frac{1}{\ell_k} G(\theta_k, z_k | \lambda_k, \lambda_{k-1}) \int_{(\theta_{k-1}, z_{k-1}) \in \Theta \times \mathcal{Z}} \left\{ \pi(d\theta_{k-1}, dz_{k-1} | \lambda_{k-1}) \right. \\ &\quad \left. \times M(\theta_{k-1}, z_{k-1}, d\theta_k, dz_k | \lambda_{k-1}) \right\}, \end{aligned}$$

where $\ell_k := L_{k-1} / L_k$ and

$$G(\theta, z | \lambda_k, \lambda_{k-1}) := \frac{L_T^N(\theta, z | \lambda_k) \psi_T^\theta(dz | \lambda_k)}{L_T^N(\theta, z | \lambda_{k-1}) \psi_T^\theta(dz | \lambda_{k-1})}.$$

This is precisely the forward recursion of a Feynman-Kac model on $\Theta \times \mathcal{Z}$ with initial distribution

$$\mathbb{M}_0 = \pi(d\theta, dz | \lambda_0),$$

transition kernels $M_k = M(d\theta_k, dz_k, \theta_{k-1}, z_{k-1} | \lambda_k)$ and potential functions $G_0 = 1$ and $G_k = G(\theta, z | \lambda_k, \lambda_{k-1})$. Hence, we have a Feynman-Kac model in which the marginals are the tempered posteriors, and the corresponding forward-recursion algorithm can be used to propagate samples across the tempering sequence. This forward-recursion constitutes the base of the SMC sampler used by Svensson et al., 2018.

Algorithm 4.1: SMC sampler for highly informative observations

Input: number of parameter samples N_θ , number of filter particles N_x , cutoff temperature $\lambda_{\min}$, observations $y_{0:T}$, initial temperature λ_0

Output: approximate samples $\{(\theta^n, z^n)\}_{n=1}^{N_\theta}$ from $\pi(d\theta, dz)$.

Operations involving index n must be performed for $n = 1, \dots, N_\theta$

```

k ← 0
(θ₀ⁿ, z₀ⁿ) ∼ π₀(dθ, dz | λ₀)      ▷ Sample using PMMH
while λₖ > λ_min do                ▷ Early stopping
  k ← k + 1
  Solve in λₖ ∈ [λ_min, λ_{k-1}] the equation:
  ESS ( { G(θ_{k-1}ⁿ, z_{k-1}ⁿ | λₖ, λ_{k-1}) }_{n=1}^{N_θ} ) = α N_θ
  If no solution exists, set λₖ ← λ_{k-1}

  wₖⁿ ← G(θ_{k-1}ⁿ, z_{k-1}ⁿ | λₖ, λ_{k-1})
  Wₖⁿ ← wₖⁿ / ∑_m wₖ^m

  Bₖ^{1:N_θ} ∼ resample(Wₖ^{1:N_θ})
  (θₖⁿ, zₖⁿ) ∼ M(θ_{k-1}^{Bₖⁿ}, z_{k-1}^{Bₖⁿ}, dθₖ, dzₖ | λₖ)
  ▷ PMMH mutation kernel
end while

```

5. Adaptive tempering

The previous discussion presupposed that we already had a tempering sequence picked out in the beginning, which will not be the case in practice. Choosing a good tempering sequence is essential for the performance of SMC sampling. When $\{\lambda_k\}_{k=0}^K$ decreases too quickly, the likelihood landscape will change rapidly, obstruct the SMC sampler, and lead to high variance; when $\{\lambda_k\}_{k=0}^K$ decreases too slowly, computation time and power will be wasted. Del Moral, Doucet, et al., 2012 argue that a good tempering sequence should keep the effective sample size approximately constant throughout the tempering sequence. As a reminder, the effective sample size (ESS) is defined as

$$\text{ESS}(\{w^n\}_{n=1}^N) = \frac{(\sum_{n=1}^N w^n)^2}{\sum_{n=1}^N (w^n)^2},$$

where $\{w^n\}_{n=1}^N$ is a set of unnormalized importance weights.

Svensson et al., 2018 hence propose the following adaptive tempering scheme: in each iteration, when we know the previous temperature λ_{k-1} , solve the equation

$$\text{ESS}(\{G(\theta_{k-1}^n, z_{k-1}^n | \lambda_k, \lambda_{k-1})\}_{n=1}^{N_\theta}) = \alpha N_\theta$$

for the next temperature $\lambda_k < \lambda_{k-1}$. Here, N_x denotes the number of particles to be used in the particle filters for likelihood estimation, N_θ denotes the number of posterior samples to propagate, and α is some fixed number in $[0, 1]$. Because all of the auxiliary variables z^n are kept track of (the particle filter densities $\psi_T^\theta(dz | \lambda_k) / \psi_T^\theta(dz | \lambda_{k-1})$ can be evaluated exactly), there is no randomness involved in this equation, and it can be solved effectively by using a numerical bisection method (Svensson et al., 2018).

When the temperature gets very low, the variance in likelihood estimation will tend to rise again (Svensson et al., 2018). For that reason, early stopping is applied: when the temperature reaches below a predefined threshold $\lambda_{\min}$, terminate the algorithm. Another useful heuristic, proposed by Del Moral, Doucet, et al., 2012, is to monitor the acceptance rates in the mutation steps and to terminate when they reach below some threshold.

6. Addendum: evaluating the particle filter densities

To implement this algorithm in practice, one would need to be able to evaluate the densities

$$\frac{\psi_T^\theta(dz|\lambda_k)}{\psi_T^\theta(dz|\lambda_{k-1})}$$

first. Svensson et al., 2018 only consider the case where the bootstrap particle filter is used, but the generalization to arbitrary Feynman-Kac models is relatively straightforward. Let $M_0^\theta, M_{1:T}^\theta, G_{0:T}^\theta$ be any Feynman-Kac model for the (tempered) state-space model in question, write $z = (x_{0:T}^{1:N_x}, a_{1:T}^{1:N_x})$, write $a_{1:T}^{1:N_x}$ for $da_{1:T}^{1:N_x}$ to avoid introducing counting measure², and define a filtration $\mathcal{F}_t = \sigma(X_{0:t}^{1:N_x}, A_{1:t}^{1:N_x})$. Then, by inspection of the sampling process in the generic particle filter algorithm (Algorithm 2.1), it can be seen that

$$\begin{aligned} \psi_T^\theta(dx_{0:T}^{1:N_x}, a_{1:T}^{1:N_x}|\lambda_k) &= \prod_{n=1}^{N_x} M_0^\theta(dx_0^n) \\ &\times \prod_{t=1}^T \prod_{n=1}^{N_x} \mathbb{P}^\theta(a_t^n|\mathcal{F}_{t-1}, \lambda_k) \\ &\times \prod_{t=1}^T \prod_{n=1}^{N_x} M_t^\theta(x_{t-1}^{a_t^n}, dx_t), \end{aligned}$$

where $\mathbb{P}^\theta(a_t^n|\mathcal{F}_{t-1}, \lambda_k)$ is the resampling distribution. For multinomial resampling, this distribution is given by the somewhat messy expression

$$\mathbb{P}^\theta(a_t^n|\mathcal{F}_{t-1}, \lambda_k) = \begin{cases} \frac{G_0^\theta(X_0^{a_t^n}|\lambda_k)}{\sum_{m=1}^{N_x} G_0^\theta(X_0^{a_t^m}|\lambda_k)}, & \text{if } t = 1; \\ \frac{G_{t-1}^\theta(X_{t-2}^{a_t^n}, X_{t-1}^{a_t^n}|\lambda_k)}{\sum_{m=1}^{N_x} G_{t-1}^\theta(X_{t-2}^{a_t^m}, X_{t-1}^{a_t^m}|\lambda_k)}, & \text{otherwise.} \end{cases}$$

This can be directly computed because $A_{1:T}^{1:N_x}, X_{1:T}^{1:N_x}$ are already observed in the filtration $\mathcal{F}_{t-1}$. When specializing to the bootstrap particle filter, the particle filter distribution reduces directly to

$$\begin{aligned} \psi_T^\theta(dx_{0:T}^{1:N_x}, a_{1:T}^{1:N_x}|\lambda_k) &= \prod_{t=1}^T \prod_{n=1}^{N_x} \frac{N(y_{t-1}|x_{t-1}^{a_t^n}, \sigma^2 + \lambda_k)}{\sum_{m=1}^{N_x} N(y_{t-1}|x_{t-1}^{a_t^m}, \sigma^2 + \lambda_k)} \\ &\times \prod_{n=1}^{N_x} P_0^\theta(dx_0^n) \prod_{t=1}^T \prod_{n=1}^{N_x} P^\theta(dx_t^n, x_{t-1}^{a_t^n}), \end{aligned}$$

which Svensson et al., 2018 had used in their implementation of the algorithm.

7. Experiments: Non-linear toy model

Svensson et al., 2018 applied their method to a non-linear state-space model given by

$$\begin{aligned} x_0 &\sim \delta_0(dx_0), \\ x_{t+1}|x_t &\sim N(\arctan(x_t) + \theta_1 u_t, 1), \quad t = 1, \dots, T \\ y_t|x_t &\sim N(|x_t| + \theta_1 \theta_2, 0.01) \quad t = 0, \dots, T, \end{aligned}$$

where u_t is a vector of exogenous input parameters. We tried to reproduce their simple parameter estimation experiment for this model to evaluate our implementation. A sample of size $T = 300$ was generated with parameters $\theta_1 = 1, \theta_2 = 1$, and fixed exogenous input parameters $u_t \sim N(0, 1)$. A prior $\nu(d(\theta_1, \theta_2)) = \text{Unif}([0, 2] \times [0, 2])$ was put on the parameters. The SMC sampler was run with $N_\theta = 300$ parameter particles and adaptive tempering parameter $\alpha = 0.4$. Mutation steps consisted of 40 PMMH iterations using a bootstrap particle filter with $N_x = 300$ particles. PMMH proposals followed a multivariate Gaussian random walk with standard deviation 0.02. Tempering was initialized at $\lambda_0 = 10$ and terminated when either $\lambda < 0.01$ or the PMMH acceptance rate fell below 0.05.

Posterior samples. The algorithm ran for $n = 666$ iterations until having reached the minimum temperature. The final particle sample obtained is shown in Figure 8.1 below.

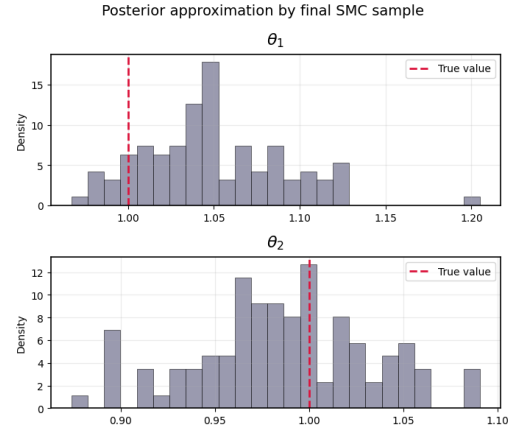


Figure 2: Result after running Algorithm 4.1 on a toy model.

This run successfully recovered the parameters with some accuracy and with relatively broad posteriors, indicating a lack of weight decay. It should be stated that additional runs have shown that the parameters are not consistently recovered, though weight decay seems to be consistently avoided. More systematic and exhaustive numerical experimentation would be insightful, but the prohibitive computational cost in this case made such experimentation infeasible using our present implementation. We have, however, performed such further experiments on an even simpler benchmark model (see Section 9).

Likelihood estimate variance. We have also investigated the variance in particle filter likelihood estimates for this same experimental run. At each iteration of the SMC sampler, the bootstrap particle filter was used to estimate the log-marginal likelihood of the sample $r = 30$ times to compute variances. The result is shown in Figure 3.

This figure indicates that the tempered SMC methodology indeed seems to solve the problem of likelihood estimator variance in the

²Following the notation in Chopin and Papaspiliopoulos, 2020

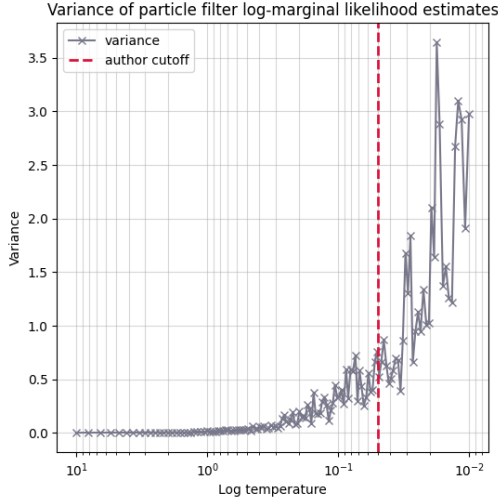


Figure 3: Log-marginal likelihood estimates versus temperatures. Red line shows the temperature cutoff used by the authors ($\lambda_{\min} = 0.05$).

mid to high-temperature range. This is to be expected, as higher temperature corresponds to broader likelihoods. What is also not surprising is the explosion of variance when the temperature gets sufficiently low. This was anticipated by the authors, and motivates the early stopping threshold. More systematic experiments into estimator variance have been performed in the next section within a simpler benchmark model.

8. Experiments: Power tempering

A widely-used method for tempering SMC samplers is *power tempering*, wherein one typically works with a tempering sequence of the form

$$\pi(d\theta|\beta) \propto \nu(d\theta) p_T^\theta(y_{0:T})^\beta$$

for $0 \leq \beta \leq 1$. As β moves from 0 to 1, the tempered posterior interpolates between the prior and the true posterior. We investigate how this method fares with highly informative observations, and compare it to the tempering scheme used by Svensson et al., 2018. Because likelihoods are not directly available, power tempering as such cannot work, but a pseudo-marginal approach may again be applied by letting the tempering sequence be

$$\pi(d\theta, dz|\beta) \propto \nu(d\theta) L_T^N(\theta, z)^\beta \psi_T^\theta(dz),$$

which has the true tempered posterior as its marginal. The PMMH kernels are modified such that the above distribution is the stationary distribution. The Feynman-Kac model for this tempering sequence comes out to be much simpler than the previous one because the particle filter no longer depends on the tempering parameter. This results in cancellation, and the potential functions become

$$G(\theta, z|\beta_k, \beta_{k-1}) = L_T^N(\theta, z)^{\beta_k - \beta_{k-1}}.$$

To benchmark power tempering against the Svensson et al., 2018 tempering scheme, we constructed a simple one-dimensional non-linear toy model, primarily to save computational resources. This model is

$$\begin{aligned} x_0 &\sim \delta_0(dx_0), \\ x_{t+1}|x_t &\sim N(x_t + \theta \arctan(x_t) + u_t, 1), \quad t = 1, \dots, T \\ y_t|x_t &\sim N(x_t, 0.001) \quad t = 0, \dots, T, \end{aligned}$$

where u_t is again some fixed vector of exogenous input variables. Several experiments were performed using this benchmark model.

RMSE across hyperparameters in the small toy model. Our first experiment investigates whether Algorithm 4.1 can recover the true parameter when using power tempering as well as it can when using Svensson et al., 2018’s tempering scheme. A sample of size $T = 200$ was generated from the state-space model under the true parameter $\theta = 1$, and exogenous inputs were generated as $u_t \sim N(0, 1)$. For both tempering schemes, Algorithm 4.1 was run with a bootstrap particle filter, parameters $N_x \in (25, 50, 100)$, $N_\theta \in (25, 50, 100)$, a $\text{Unif}([-5, 5])$ -prior on the parameter, adaptive tempering parameter $\alpha = 0.5$ and random walk Gaussian PMMH kernels with adaptive standard deviation and 25 iterations. Standard deviations were adapted such that, in each iteration, it was set to $\sigma_k = 2.38 \cdot \widehat{\text{SE}}(\theta_{k-1}^n)$, in accordance with a well-known optimal scaling result for Metropolis-Hastings sampling (Roberts et al., 1997). For power tempering, the initial temperature was set to $\beta = 0$, and for the other tempering scheme, the initial temperature was set to $\lambda = 10$. Each run was repeated $r = 10$ times.

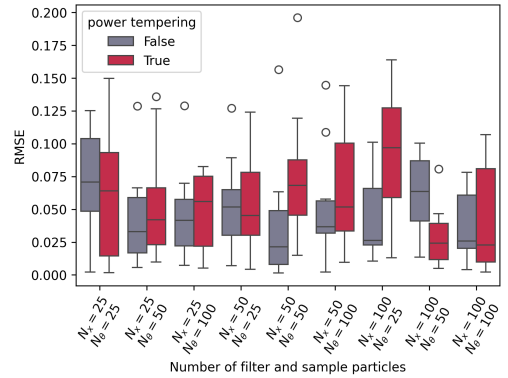


Figure 4: Box-plots of the RMSE (root mean square error) computed between final samples and true parameter after 10 repeats per hyperparameter combination.

The RMSE (Root Mean Square Error) was calculated between each set of final samples and the true parameter. These results give an early indication that power tempering may be competitive with Svensson et al., 2018’s tempering scheme for parameter learning with highly informative observations.

RMSE across hyperparameters in the larger toy model. We have also evaluated the power tempering scheme on the nonlinear toy model from Svensson et al., 2018, though without comparison to the proposed tempering scheme. A sample of size $T = 200$ was generated from the state-space model under the true parameter $\theta = (1, 1)$, and exogenous inputs were generated as $u_t \sim N(0, 1)$. Algorithm 4.1 was run with a bootstrap particle filter, parameters $N_x \in (50, 150, 300)$, $N_\theta \in (50, 150, 300)$, a $\text{Unif}([0, 2] \times [0, 2])$ -prior on the parameter, adaptive tempering parameter $\alpha = 0.5$ and random walk Gaussian PMMH kernels with adaptive standard deviation and 25 iterations. Standard deviations were adapted such that, in each iteration, it was set to $\sigma_k = 2.38 \cdot \widehat{\text{SE}}(\theta_{k-1}^n)$. The initial temperature was set to $\beta = 0$. Each run was repeated $r = 10$ times.

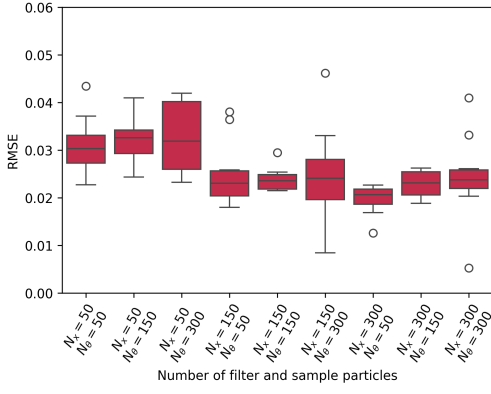


Figure 5: Box-plots of the RMSE (root mean square error) computed between final samples and true parameter after 10 repeats per hyperparameter combination.

These results indicate that power tempering also performs well at parameter learning in the more difficult nonlinear toy model.

Number of iterations. Iteration counts were recorded for each run in the previous two experiments, and median iteration counts were computed over repeats per hyperparameter combination. Ratios of median iteration counts (Svensson tempering / power tempering) were computed and plotted in Figure 6.

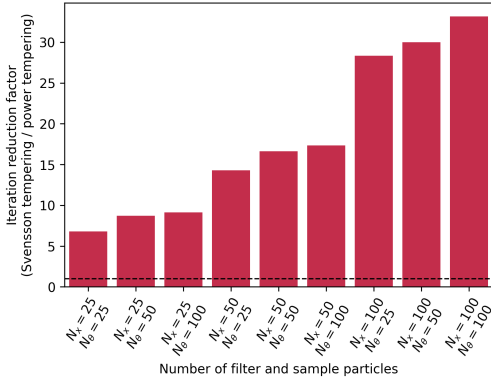


Figure 6: Ratios between median iteration counts across repeats for Svensson tempering and power tempering.

This experiment indicates a remarkable reduction in iteration counts when using power tempering instead of Svensson et al., 2018’s proposed tempering scheme, especially as the number of particles grows larger, all the while parameter estimation results for these hyperparameter combinations remain competitive with the proposed algorithm.

Furthermore, Figure 7 shows median iteration counts for the experimental power tempering runs in the larger non-linear toy model. Again, remarkably low iteration counts are recorded, all the while the parameter estimates are accurate. Although we have not compared power tempering with Svensson et al., 2018’s tempering scheme on this model, an experiment in Svensson et al., 2018 concerning expected iteration counts under this same toy model indicates that their method is expected to take around 100-200 iterations before reaching their chosen cutoff temperature (that being $\lambda_{\min} = 0.05$ in their experiment).

Likelihood estimate variances. The data generated in the RMSE ex-

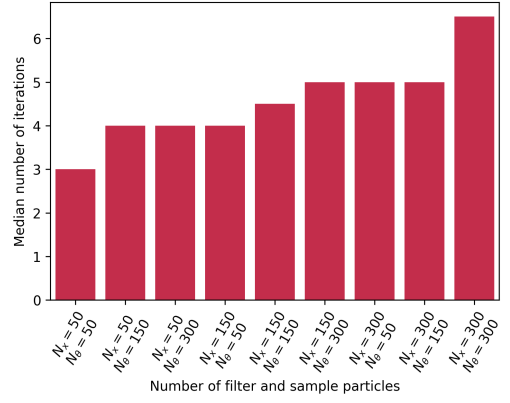


Figure 7: Median iteration counts across repeats for power tempering.

periment was used to assess likelihood estimator variances across hyperparameter combinations. For each assessed hyperparameter combination, repeat, and temperature, a bootstrap particle filter was run $r = 10$ times to estimate the variance of the log-marginal likelihood of the observations. In Figure 8, three randomly selected trajectories of the log-marginal likelihood variance are plotted for each of the two tempering schemes, to give a rough indication of their behaviors.

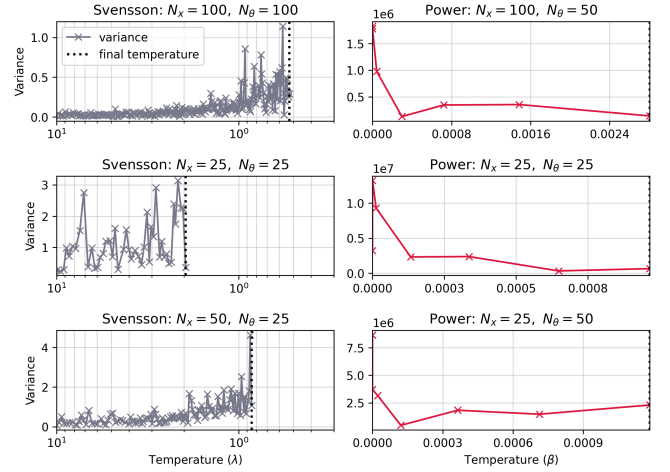


Figure 8: For each of the two tempering schemes, three random log-marginal likelihood variance trajectories over temperature progression.

In Figure 8, it can be seen that power tempering barely reduces likelihood estimation variance. This is not merely a characteristic of these three randomly selected trajectories: in fact, as Table 1 shows, likelihood estimator variance is extremely high under power tempering in all regimes, whereas the proposed tempering scheme succeeds in controlling variance.

Table 1: Range of likelihood-estimator variances across all temperatures, repeats, and hyperparameter combinations.

Tempering scheme	Min	Median	Max
Power	7.16×10^4	1.11×10^6	3.97×10^7
Svensson	2.46×10^{-3}	2.11×10^{-1}	5.04×10^0

This can be explained by the fact that power tempering does not

change anything in the particle filter likelihood estimation, and hence all estimates are made under the highly informative model. Another point of note in Figure 6 is the very tiny temperature difference before the algorithm comes to a halt while using power tempering. This apparently suffices for locating the true parameter. We might speculate that power tempering still works, then, in spite of its noisy likelihood estimates because small changes in temperature are sufficient to induce enough selection pressure among particles to distinguish between plausible and implausible parameter values.

This observation raises an interesting question regarding the role of likelihood estimator variance in SMC-based parameter learning. Svensson et al., 2018, and indeed, conventional wisdom regarding weight degeneracy, suggest that accurate likelihood estimates are essential, motivating tempering schemes specifically designed to control their variance. However, the results presented here indicate that, at least in this simple setting, successful parameter learning may not require precise likelihood estimation. Power tempering appears able to exploit the strong identifiability of the model, achieving competitive estimation accuracy despite likelihood estimator variance being extremely high.

9. Discussion

The experiments presented in this review suggest that the tempering scheme proposed by Svensson et al., 2018 successfully addresses the specific problem it was designed for. Across all experiments, tempering through the addition of noise to the observation variance indeed reduced the variance of particle filter likelihood estimates, doing so by several orders of magnitude relative to classical power tempering. In the low temperature regime, however, variance still tends to spike, and the incorporation of further variance reduction techniques, like the bridging particle filter (Del Moral and Murray, 2015) or the alive particle filter (Moral et al., 2015) like could be of great benefit (these particular examples were pointed out by Svensson et al., 2018).

The experiments reveal a somewhat unexpected result. Despite producing highly variable likelihood estimates, power tempering often achieved parameter estimates that were competitive with those obtained using the proposed tempering scheme. Furthermore, power tempering generally required significantly fewer tempering iterations, and hence a much lower computational cost. This suggests that accurate likelihood estimation and successful parameter learning may not be as tightly coupled as is commonly assumed, at least in the simple benchmark models considered here.

One possible explanation is that, when the model is strongly identifiable, even noisy likelihood estimates may contain sufficient information to distinguish plausible parameter values from implausible ones. This raises interesting questions regarding the extent to which variance reduction in likelihood estimation is actually necessary for successful parameter inference.

These observations should be interpreted with caution. The experiments performed here are limited in scope and rely primarily on low-dimensional toy models. More complex, high-dimensional state-space models may exhibit different behaviour. A natural direction for future work would be to investigate whether the apparent robustness of power tempering persists in more challenging applications. For example, one could experiment with power tempering on the Wiener-Hammerstein benchmark (Ljung et al., 2009) on which Svensson et

al., 2018 also benchmarked their own method.

Overall, the results support the effectiveness of the methodology proposed by Svensson et al. for controlling likelihood estimator variance in highly informative state-space models. However, they also suggest that alternative tempering strategies may remain viable for parameter learning even when likelihood estimates are extremely noisy, highlighting an interesting avenue for further research.

10. Code availability

The code used for the implementation of the algorithms in Svensson et al., 2018, as well as for all simulations and experiments reported in this review, is publicly available at <https://github.com/vvninif/smc-highly-informative-observations>.

References

- Chopin, Nicolas, Pierre E. Jacob, and Omros Papaspiliopoulos (2012). *SMC²: an efficient algorithm for sequential analysis of state space models*. arXiv: 1101.1528 [stat.CO]. URL: <https://arxiv.org/abs/1101.1528>.
- Chopin, Nicolas and Omros Papaspiliopoulos (2020). *An introduction to Sequential Monte Carlo / Nicolas Chopin, Omros Papaspiliopoulos*. eng. Springer series in statistics. Cham, Switzerland: Springer. ISBN: 9783030478452. URL: <http://liverpool.idm.oclc.org/login?url=http://link.springer.com/10.1007/978-3-030-47845-2>.
- Del Moral, Pierre, Arnaud Doucet, and Ajay Jasra (2012). “An Adaptive Sequential Monte Carlo Method for Approximate Bayesian Computation”. In: *Statistics and Computing* 22.5, pp. 1009–1020. ISSN: 1573-1375. DOI: 10.1007/s11222-011-9271-y. URL: <https://doi.org/10.1007/s11222-011-9271-y>.
- Del Moral, Pierre and Lawrence M. Murray (2015). “Sequential Monte Carlo with Highly Informative Observations”. In: *SIAM/ASA Journal on Uncertainty Quantification* 3.1, pp. 969–997. DOI: 10.1137/15M1011214. eprint: <https://doi.org/10.1137/15M1011214>. URL: <https://doi.org/10.1137/15M1011214>.
- Kantas, Nikolas et al. (Jan. 2015). “On particle methods for parameter estimation in state-space models”. English. In: *Statistical Science* 30.3. Publisher Copyright: © 2015 by the Institute of Mathematical Statistics., pp. 328–351. ISSN: 0883-4237. DOI: 10.1214/14-STS511.
- Ljung, Lennart, Johan Schoukens, and Johan Suykens (July 2009). “Wiener-Hammerstein Benchmark”. In: *Proceedings of the 15th IFAC Symposium on System Identification*. Presented at the 15th IFAC Symposium on System Identification. Saint-Malo, France. URL: <https://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-89070>.
- Moral, Pierre Del et al. (2015). “The alive particle filter and its use in particle Markov chain Monte Carlo”. In: *Stochastic Analysis and Applications* 33.6, pp. 943–974.
- Roberts, G. O., A. Gelman, and W. R. Gilks (1997). “Weak Convergence and Optimal Scaling of Random Walk Metropolis Algorithms”. In: *The Annals of Applied Probability* 7.1, pp. 110–120. ISSN: 10505164. URL: <http://www.jstor.org/stable/2245134> (visited on 06/03/2026).

Svensson, Andreas, Thomas B. Schön, and Fredrik Lindsten (2018).
“Learning of state-space models with highly informative observations: a tempered Sequential Monte Carlo solution”. In: *Mechanical Systems and Signal Processing* 104, pp. 915–928.